

Article

A Global-Speed-Region Voltage Angle Control Method Considering Dead-Time Effect Compensation for Permanent Magnet Synchronous Motor Drives

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Abstract: The voltage utilization in the PMSM control can be improved through the voltage angle control (VAC) method, thereby enhancing the control efficiency and output power capability. However, the traditional VAC method can only be applied to a high-speed region, and it is still necessary to switch to another control mode in the low-speed region, resulting in a significant torque jitter during the switching process. Meanwhile, the deviation between the actual current operating point and the optimal one is caused by the dead-time effect in the traditional VAC method, causing more energy loss. To solve these problems, a global-speed-region VAC method considering dead-time effect compensation is proposed in this article. Firstly, the application area of the proposed method is extended to both the high-speed and low-speed region on the basis of analyzing the variation function of the modulation index (MI), achieving global non-mode-switching control and eliminating the torque jitter. Additionally, the relationship between the dead time and MI is quantitatively analyzed, and a global-speed-region VAC method is proposed to compensate the dead-time effect and reduce the energy loss. Finally, the effectiveness of the proposed method is verified in PMSM bench experiments.



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Keywords: PMSM; voltage angle control; torque jitter; dead-time effect

1. Introduction

Due to their high control efficiency and power density, permanent magnet synchronous motors (PMSMs) are widely used in the battery electric vehicle (BEV) fields and other industrial fields [1–4]. The field-weakening (FW) control method is applied to plan the current operating point because the increasing back electromotive force (BEMF) needs to be overcome by a designed negative d -axis current [5,6]. However, PMSMs suffer from the problems of voltage saturation and cross coupling in the high-speed region [7]. This limits the improvement in the voltage utilization, resulting in a negative impact on control efficiency and output power capability. In the VAC method, the control of voltage amplitude and voltage angle belongs to a decoupled architecture, which can avoid the above two problems [8,9]. So, the higher voltage utilization can be achieved, which produces a better performance on the control efficiency and output power capability [10,11].

The VAC methods can be divided into two categories, one is the single-current-regulator (SCR) method [12,13], and the other is the non-current-regulator (NCR) method [14,15]. A typical SCR method called the single q -axis current regulator-variable voltage angle (SQCR-VVA) method is proposed in [12]. In this method, the reference value of the voltage angle is obtained through the q -axis current loop, and the stable control can only be achieved on the right part of the characteristic current, not on the left part. The single d -axis current regulator-regulations q -axis voltage (SDCR-RQV) method is another

typical SCR method [13]. In this method, the reference value of the q -axis voltage is obtained through the d -axis current loop, and its stable control region is extended to the left part of the characteristic current. But the symbol modification is required when crossing the different parts. Ref. [14] proposed a typical NCR method. There is no current loop in this method, and the reference value of the voltage angle is obtained through the torque loop. The NCR method has the advantages of simpler control architecture and better dynamic torque performance [15]. However, the traditional VAC methods can only be applied to the high-speed region, and it is necessary to switch to another control mode in the low-speed region, resulting in a significant torque jitter during the switching process [13,15]. The comparison of existing VAC methods is shown in Table 1. It can be seen that all existing VAC methods can only be applied in the high-speed region. So, it is important to develop a novel VAC method that can be applied in both high-speed and low-speed regions.

Table 1. Comparison of existing VAC methods.

	Methods	Voltage Utilization	Application Area
SCR	Constant voltage method	<1.00	high-speed region, $n > n_0$
	Variable voltage method	1.00	high-speed region, $n > n_0$
	Voltage Vector Phase Angle Method	1.00	high-speed region, $n > n_0$
NCR	-	1.00	high-speed region, $n > n_0$

Meanwhile, the ability of closed-loop regulation for the voltage amplitude control cannot be obtained due to its feedforward structure in the VAC method [14]. So, the influence of the dead-time effect on the voltage amplitude cannot be suppressed, which leads to the deviation between the actual current operating point and the optimal one, and further causes more energy loss. So, the compensation for the dead-time effect is very important. According to the different way of being implemented, the dead-time effect compensation methods are mainly divided into the following three types:

The first type is hardware compensation methods. Ref. [16] proposed a voltage feed-forward compensation method, which directly compensates the dead-time error voltage through the $\alpha\beta$ currents. It is simple and easy to be applied in engineering, but requires additional detection circuits. The moving average method proposed by [17] optimized the algorithm calculation logic, but it also requires additional detection circuits, which increases both cost and system complexity. The method proposed by [18] filtered the current component based on EKF, which reduces the difficulty of determining the polarity of the current to some extent. However, it also requires additional detection circuits and shares a limited compensation performance. All hardware compensation methods require the additional detection circuits, which not only increase the hardware costs but also introduce a control cycle delay between the measured signal and the ideal signal, leading to measurement errors and affecting the effectiveness of compensation.

The second type is software compensation methods. The method proposed in [19] assists in determining the polarity of current polarity based on the change rate in the phase current, reducing the impact of the current clamping effect, but there is a lack of analysis on the total harmonic distortion. Ref. [20] proposed the trapezoidal compensation method, which can be applied to the overmodulation region, but it requires more accurate current polarity judgment. Ref. [21] proposed a new SPWM modulation strategy to optimize the compensation performance, but this method also relies on accurate current polarity judgment. The software compensation methods do not require additional hardware circuits, and the compensation performance is relatively accurate, but generally requires accurate current polarity judgment.

The third type is the other compensation methods. A high-gain feedback control method was proposed in [22]. The tracking characteristics between the actual and target voltage are achieved through a voltage outer loop, and the dynamic performance of the system is achieved through a current inner loop. The method does not rely on accurate

current polarity judgment, but it is difficult to be applied when the AC frequency changes such as in the motor drive system for electric vehicles. Ref. [23] proposed an observer estimation method, which uses a voltage model to estimate and compensate the dead-time error voltage. It also does not require accurate current polarity judgment, but the algorithm is complex, resulting in a large computational load. The comparison of the three types of dead-time effect compensation methods mentioned above is shown in Table 2. Overall, the existing dead-time effect compensation methods cannot simultaneously meet the following requirements: (1) no need for additional hardware; (2) no need for accurate current polarity judgment; (3) can be applied when the AC frequency changes; (4) small computational load.

Table 2. Comparison of the dead-time effect compensation methods.

Methods		Advantage	Disadvantage
Other method	Hardware compensation methods	smaller computational load	additional hardware costs, measurement errors
	Software compensation methods	easy to be implemented	relying on accurate current polarity judgment
	High gain feedback control method	no need for accurate current polarity judgment	difficult to be applied when the AC frequency changes
	Observer estimation method	no need for accurate current polarity judgment	relying on accurate motor parameters; big computational load

To solve the above problems, a global-speed-region VAC method considering dead-time effect compensation for PMSM drives is proposed. Firstly, the variation function in the modulation index in the low-speed region is obtained, and on this basis, the application area of VAC is extended to the global speed region. In addition, a quantitative relationship between the dead time and the modulation index is analyzed, and a new compensation method for the dead-time effect based on modulation index transformation is proposed. The main contributions of this article are as follows:

- (1) A global-speed-region VAC method. In this article, a global-speed-region method is proposed, which expands the application area of the VAC method to the entire speed region, and solves the problem of torque jitter generated by the traditional VAC method during the transition between low-speed and high-speed ranges;
- (2) A new compensation method for the dead-time effect. In this article, a new dead-time effect compensation method is proposed, based on the modulation index transformation. This method has the advantage of not requiring additional hardware circuits compared to the hardware compensation methods, not relying on accurate current polarity judgment compared to the software compensation methods, and being applicable to situations with AC frequency changes and requiring less computation compared to the high-gain feedback control method and observer estimation method. More importantly, the current operating points are returned to the optimal points, reducing the additional energy loss caused by the dead-time effect.

The rest of this article is organized as follows. Section 2 describes the traditional VAC method and its limitations. The global-speed-region VAC method is analyzed in Section 3, and the dead-time effect compensation method based on *MI* compensation in Section 4. In Section 5, the effectiveness of the proposed method is verified by bench experiments. Finally, this paper is concluded in Section 6.

2. Traditional VAC Method and Its Limitations

There are two limitations in the traditional VAC methods. One is the control-mode-switch during the transition between low-speed and high-speed ranges, which causes the torque jitter. Another is the control error of *MI* caused by the dead-time effect, which causes the current operating points deviate from the optimal points. The structure of the NCR method is more concise and the control of torque is more direct, so in this paper, research and verification is conducted based on the NCR method. The system block diagram of NCR method is shown in Figure 1.

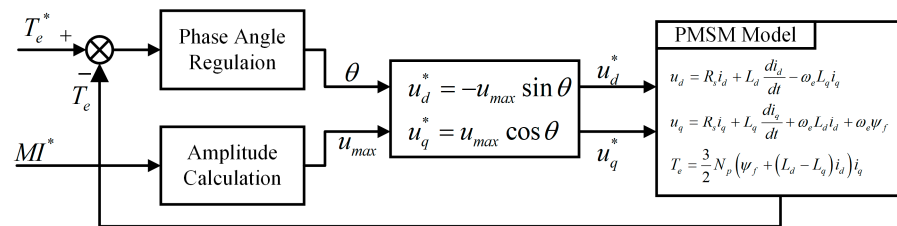


Figure 1. The system block diagram of NCR.

2.1. The Model of the NCR Method

The relationship between the voltage angle and the dq -axes current is as follows, and Appendix A refers to the derivation process:

$$i_d = \frac{u_{max} \cos \theta - \omega_e \psi_f}{\omega_e L_d} \quad (1)$$

$$i_q = \frac{u_{max} \sin \theta}{\omega_e L_q} \quad (2)$$

where i_d and i_q are the dq -axes current; L_d and L_q are the dq -axes inductance; ω_e is the electrical rotation speed; ψ_f is the flux linkage; u_{max} is the maximal voltage amplitude; and θ is the voltage angle between the voltage vector and the q -axis.

And the relationship between the voltage angle and the torque can be obtained by substituting (1)–(2) into (A3), as follows:

$$T_e = \frac{3N_p u_{max}}{2\omega_e^2 L_d L_q} \left((L_d - L_q) u_{max} \cos \theta + L_q \omega_e \psi_f \right) \sin \theta \quad (3)$$

The monotonicity between T_e and θ is proved in [14] according to (3), which provides the theoretical fundament for the controllability of NCR.

2.2. Limitations of the Control Mode Switching

The working area of the PMSM can be divided into two parts. In the low-speed range, the current operating point falls on the MTPA curve, as shown by the curve OB in Figure 2. In the high-speed range, the current operating point falls within the FW region, as shown in the area enclosed by the OBCD area in Figure 2.

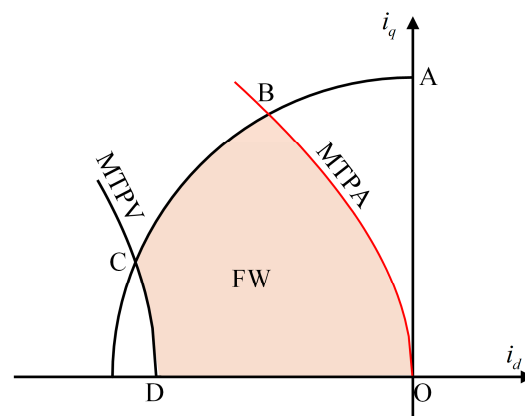


Figure 2. Working area of the PMSM (MTPA region and FW region).

However, traditional VAC methods can only be applied within the FW region, and the dual-current-regulator (DCR) method is required in the MTPA region. So, the control mode switching is necessary when the motor enters the high-speed region from the low-speed region. The criteria for determining mode switching in traditional VAC methods are shown

in Figure 3. When the reference value of the voltage amplitude exceeds the maximum value, the motor enters the FW region and the control mode switches to the VAC method. And when the actual d -axis current is less than that on the MTPA curve corresponding to the target torque curve, the motor enters the MTPA region and the control mode switches to the DCR method. Torque jitter is inevitable during the switching process. More seriously, if the current operating point falls on the boundary between the FW and the MTPA region, the switching of control modes will be particularly frequent, causing extreme torque jitter.

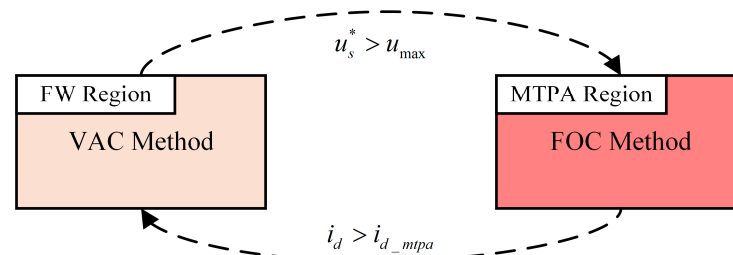


Figure 3. The criteria for determining mode switching in traditional VAC methods.

2.3. Limitations of the Dead-Time Effect

It can be easily obtained from Figure 1 that the optimal current operating point is determined by the target torque T_e^* and the target modulation index MI^* in the traditional VAC method. In other words, the optimal current operating point falls at the intersection of the torque curve and the voltage ellipse, as shown in Figure 4, the optimal current operating points E and F are the intersections of the ETC and the VLE1 and VLE 2.

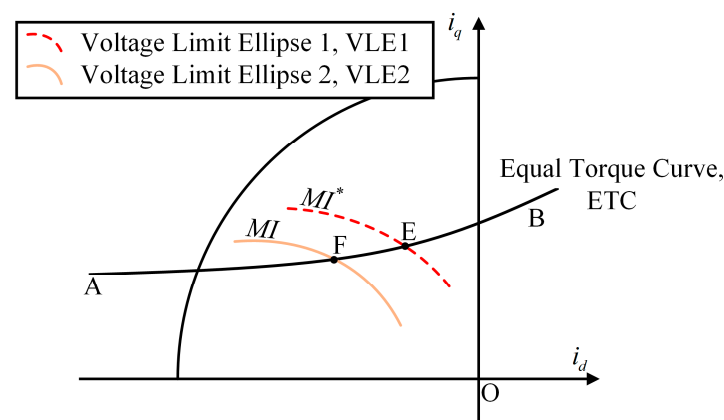


Figure 4. The actual current operating point and the optimal current operating point.

However, the closed-loop regulation capability of MI is not obtained in traditional VAC methods. The control error between the actual MI and the target MI^* is generated due to the dead-time effect. This causes the actual current operating points to deviate from the optimal ones. As shown in Figure 4, the ETC corresponding to T_e^* is represented by the curve AB, the voltage limit ellipse corresponding to MI^* is represented by the VLE1, and that corresponding to the actual MI is represented by the VLE2. The optimal current operating point is represented by the intersection point E between curve AB and VLE1, while the actual current operating point is represented by the intersection point F between ETC and VLE2. It can be seen that there is a deviation between the actual and the optimal current operating point due to the control error of the actual MI and the target MI^* .

3. Global-Speed-Region VAC Method

Before the motor enters the FW region, the speed is low and the optimal current operating point is on the MTPA curve. The MI of the current operating point on the MTPA

curve varies with different speeds, while the target MI of the traditional VAC method is constant. Therefore, it cannot obtain the optimal current operating point in low-speed regions. The change rule of the MI on the MTPA curve has not been studied. In this paper, the functional relationship between the MI and the motor speed on the MTPA curve is firstly derived, and based on this, in this paper, a global-speed-region VAC method is proposed for solving that problem of the traditional VAC method.

3.1. Theory of the Global-Speed-Region VAC Method

According to Appendix B, the modulation index of the current point on the MTPA curve MI_{mtpa} can be obtained, as follows:

$$MI_{mtpa} = y(T_e, \omega_e) = \frac{\sqrt{(\omega_e L_q g(T_e))^2 + (\omega_e L_d f(T_e) + \omega_e \psi_f)^2}}{\frac{2}{\pi} u_{dc}} \quad (4)$$

Obviously, it is very difficult to directly calculate the analytical expressions of the modulation index, torque, and speed through the above equation. In this article, a surface graph of the relationship between them is presented using the offline numerical solution method, as shown in Figure 5. The red dashed line in the figure represents the base speed curve. The left area of the base speed curve with lower speed is the MTPA region, and the optimal current operating point is the intersection of the MTPA curve and the ETC curve; the area on the right side of the high-speed base speed curve is the FW region, and the optimal current operating point is the intersection of the VLE and ETC curves. From Figure 5, it can be seen that the traditional VAC method has a constant MI in both the FW and MTPA regions, so it cannot guarantee the optimal operating point in the MTPA region; the MI in the global-speed-region VAC method will vary with torque and speed in the MTPA region, ensuring that the current operating point always remains at the intersection of the MTPA and the ETC curve. Therefore, the optimal current operating point can be obtained in both the FW and MTPA regions.

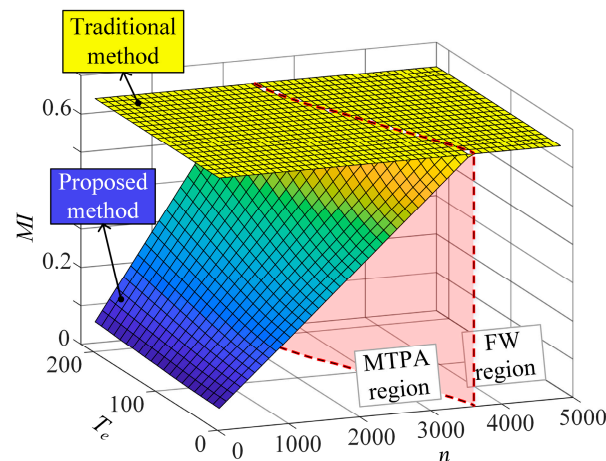


Figure 5. Curve diagram of the relationship between MI , T_e , and n .

In summary, the global-speed-region VAC method extends the controllable speed range to the full speed range including FW and MTPA regions, as shown in Figure 6.

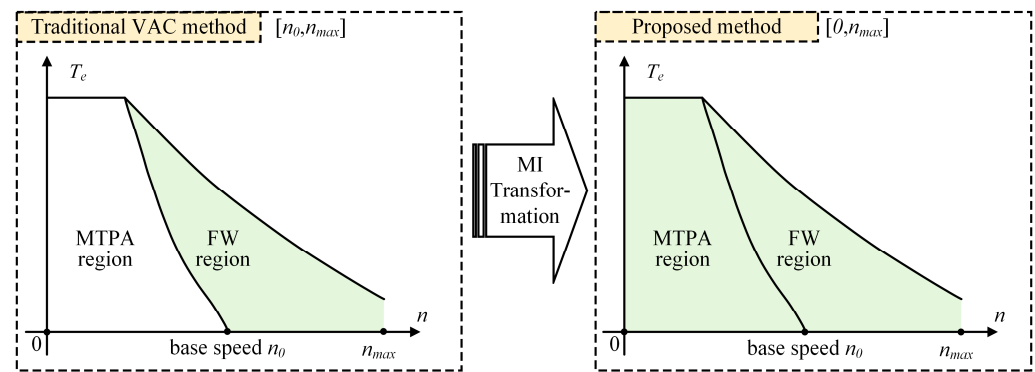


Figure 6. The speed controllable range of the traditional VAC method and the proposed method.

3.2. Control Block of the Global-Speed-Region VAC Method

The control block diagram of the global-speed-region VAC method is shown in Figure 7. Firstly, the actual torque and speed are obtained through the torque and speed measurement/calculation module; then, the MI_{mtpa} is obtained through the modulation index transformation module. If MI_{mtpa} is smaller than the reference MI^* , the motor is in the MTPA region, and the intermediate modulation index MI^{*f} is taken as MI_{mtpa} . If MI_{mtpa} is bigger than or equal to the reference MI^* , the motor is in the FW region, and the intermediate modulation index MI^{*f} is taken as MI^* . The subsequent control diagram is the same as the traditional VAC method, that is, the phase angle of the voltage vector is obtained through a regulator for the torque, which is then combined with MI^{*f} to calculate the target dq -axis voltage.

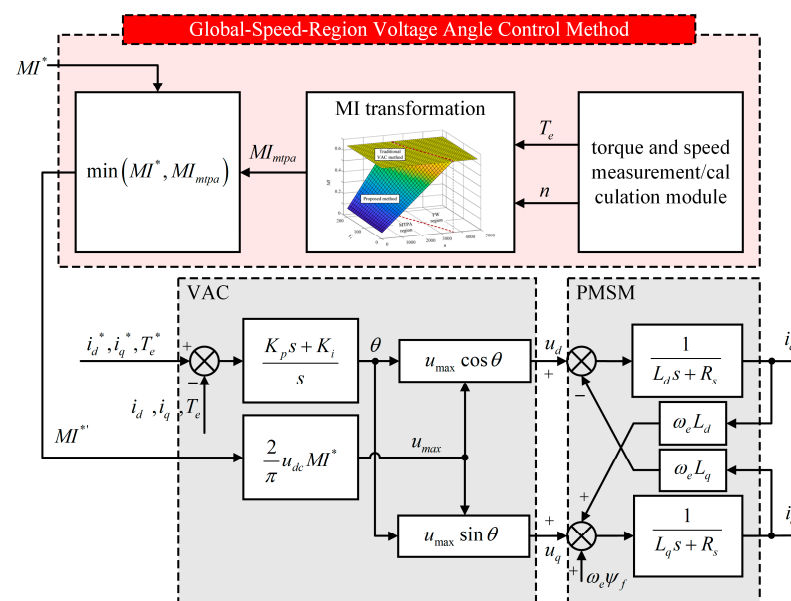


Figure 7. Control block of the global-speed-region VAC method.

4. Dead-Time Effect Compensation Method Based on MI Compensation

The dead time effect on the phase voltage is directly related to the polarity of the current i_a [16]. When $i_a > 0$, voltage error $\Delta u = -u_{dc}T_d/T_c$, and when $i_a < 0$, $\Delta u = u_{dc}T_d/T_c$, where T_c is the PWM period. The relationship between the average voltage error ΔU and the direction of current within one electrical cycle time T is shown in Figure 8, where $T = NT_c$, N is the carrier ratio. The calculation formula for ΔU is as follows:

$$\Delta U = \text{sign}(i) \frac{NT_d}{T} u_{dc} \quad (5)$$

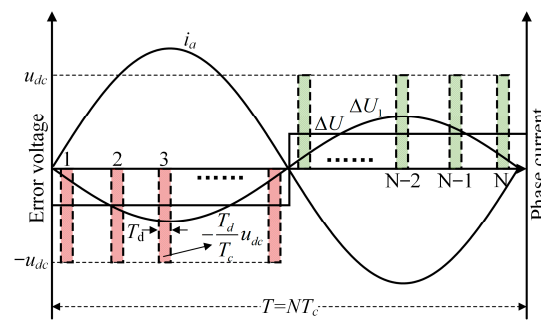


Figure 8. The average voltage error in one fundamental period and its corresponding relationship with current direction.

According to the Fourier expansion of the square wave, we can calculate the effective value of the average voltage error ΔU_1 , which is opposite in direction to the current.

$$\Delta U_1 = \frac{2\sqrt{2}}{\pi} \Delta U \quad (6)$$

The actual voltage vector $\mathbf{U}_1$, reference voltage vector $\mathbf{U}_1^*$, actual current vector $\mathbf{i}$, and voltage error vector $\Delta \mathbf{U}_1$ are shown in Figure 9. Among them, there is a 180° difference between the direction of $\mathbf{i}$ and $\Delta \mathbf{U}_1$, and the angle between $\mathbf{U}_1$ and $\mathbf{i}$ is the power factor angle φ . According to the cosine theorem, the relationship between those three voltage vectors can be obtained, as follows:

$$|\mathbf{U}_1^*|^2 = |\mathbf{U}_1|^2 + |\Delta \mathbf{U}_1|^2 - 2|\mathbf{U}_1|^2 |\Delta \mathbf{U}_1|^2 \cos(\pi - \varphi) \quad (7)$$

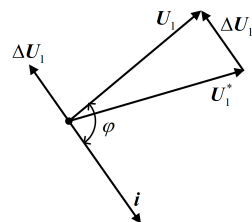


Figure 9. Superposition diagram of voltage vectors.

In SVPWM, the relationships between $|\mathbf{U}_1^*|$, $|\mathbf{U}_1|$ and MI^* , MI are as follows:

$$|\mathbf{U}_1^*| = \frac{2}{\pi} u_{dc} MI^* \quad (8)$$

$$|\mathbf{U}_1| = \frac{2}{\pi} u_{dc} MI \quad (9)$$

The quantitative relationship between the actual modulation index MI and the reference modulation index MI^* can be obtained by substituting (5)–(6) and (8)–(9) into (7).

$$MI = \sqrt{MI^{*2} - K^2 \sin^2 \varphi} - K \cos \varphi \quad (10)$$

where $K = \frac{8NT_d}{\pi T}$. Note that $N/T = Nf_e = f_c$, where f_e is the electric frequency, and f_c is the switching frequency. And the relationship between MI , MI^* , and φ is shown in Figure 10, where the dead-time T_d is set to $3 \mu\text{s}$, and the switching frequency f_c is set to 10 kHz .

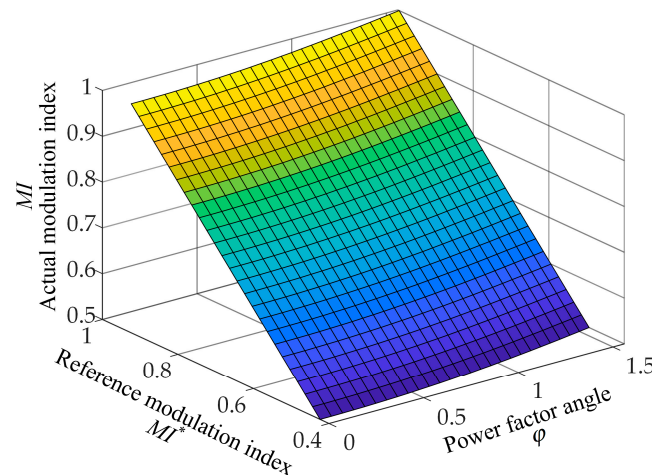


Figure 10. The relationship between MI , MI^* , and φ .

The inverse function of Formula (10) is as follows:

$$MI^* = \sqrt{(MI + K \cos \varphi)^2 + K^2 \sin^2 \varphi} \quad (11)$$

Substitute the reference modulation index MI^* as the independent variable into Equation (11) to obtain an intermediate modulation index MI_{int} . This inverse function provided the theoretical foundation for the proposed dead-time effect compensation method. The system block diagram d is shown in Figure 11.

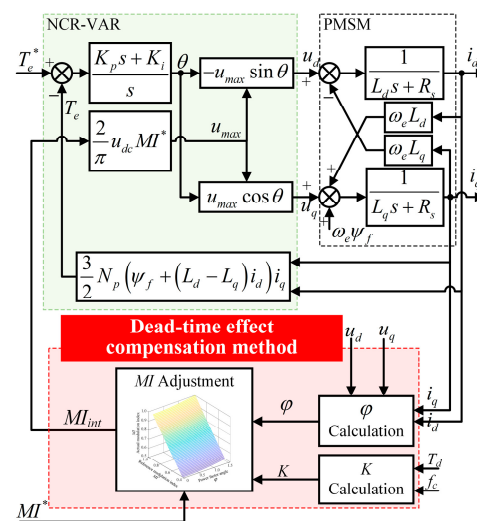


Figure 11. The system block diagram of the proposed dead-time effect compensation method.

Overall, the advantages of the dead-time effect compensation method proposed in this paper include the following: (a) no need for additional hardware circuits; (b) not relying on accurate current polarity judgment; (c) can be applied when the AC frequency changes; (d) the computational workload is relatively small.

5. Experimental Results

The general system block diagram in this paper (global-speed-region VAC + dead-time effect compensation method) is shown in Figure 12. Firstly, MI^{*f} is obtained from MI^* through the global-speed-region VAC module. Then, MI_{int} is obtained from MI^{*f} through the dead-time effect compensation module. Finally, MI_{int} is input as the reference modulation index into the VAC control architecture.

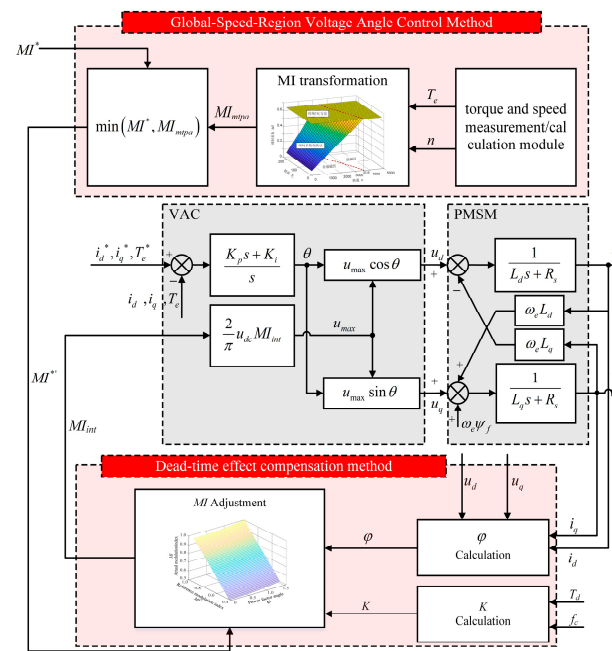


Figure 12. The general system block diagram of this paper (global-speed-region VAC + dead-time effect compensation method).

Two problems have been solved by the method proposed in this paper, one is the problem of method switching in the traditional VAC method, and the other is the problem of dead-time effect causing the current operating points to deviate from the optimal ones. In order to verify the effectiveness of the method proposed in this article in the above two aspects, a PMSM towing experimental platform is built, which is shown in Figure 13, and the parameters of the tested motor are shown in Table 3. A three-phase two-level voltage source IGBT inverter is used as the controller, with a switching frequency of 10 kHz and a battery voltage of 300 V. The percentage of dead time to PWM cycle is 3%. The controller chip is Infineon SAK-TC234L-32F200F, and its main frequency can reach 200 MHz. The phase current and voltage are sampled by the LEM LF 1010-S sensor. The experimental data are recorded through CAN.

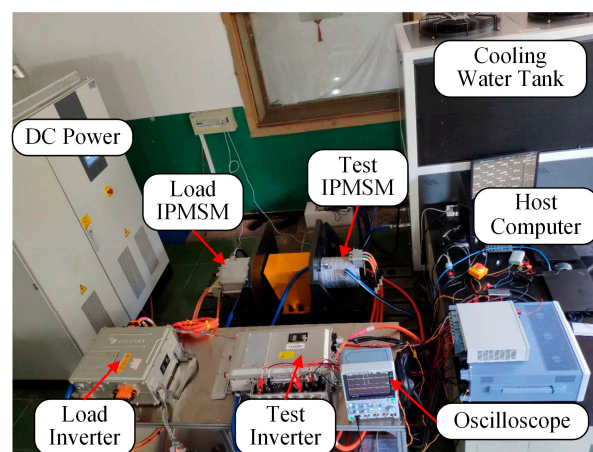


Figure 13. PMSM towing test bench.

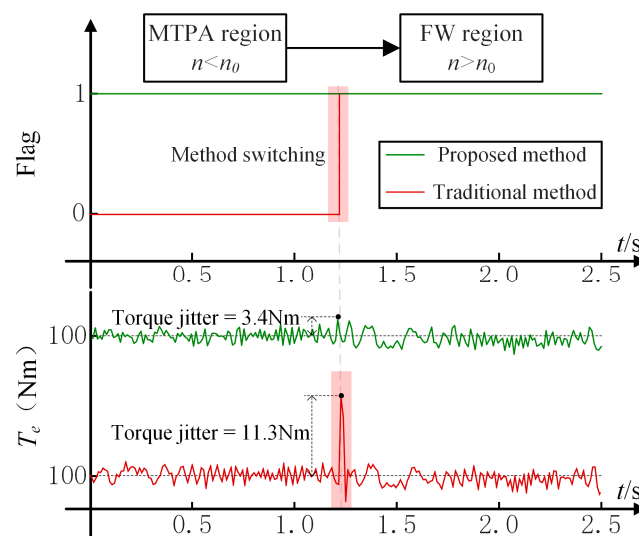
Table 3. Parameters of the tested PMSM.

Parameter	Value	Parameter	Value
DC bus voltage (v)	300	Pole pairs	4
Rated power (kW)	30	D-axes inductance (mH)	0.25
Peak power (kW)	60	Q-axes inductance (mH)	0.65
Rated torque (Nm)	100	Stator resistance (Ω)	0.02
Peak torque (kW)	200	Flux linkage (Wb)	0.072

5.1. Experimental Results of Global-Speed-Region VAC Method

In order to verify whether the problem of method switching can be solved by the global-speed-region VAC method, two sets of experiments were designed. The tested motor is set to torque mode, with target torques of 100 Nm and 200 Nm, respectively. The towing motor is set to speed mode. The modulation index increases with the increase in motor speed.

When the target torque is 100 Nm, the base speed $n_0 = 3520$ rpm, and the experimental results of the traditional and the proposed methods are shown in Figure 14. The red and green lines represent the experimental results of the traditional method and the method proposed in this paper, respectively. The experimental waveforms near the base speed are represented by the red areas. The left side of the base speed is the MTPA region, and the right side is the FW region. The traditional VAC method can only be applied in the FW region, so when the $n > 3520$ rpm, the control method needs to be switched. From the torque waveform in red line, it can be seen that the motor torque showed significant jitter at the moment of method switching, with a maximum value of 11.3 Nm and a jitter amplitude accounting for 11.3% of the target torque. While the proposed method can be applied in both the MTPA region and the FW region, so there is no method switching. In the experimental results of the proposed method, the flag remained constant at 1, and the torque waveform showed minimal jitter, with the maximum jitter being only 3.4 Nm.


Figure 14. The controllable range of traditional methods and the proposed method (100 Nm).

When the target torque is 200 Nm, the base speed $n_0 = 2404$ rpm, and the experimental results of the traditional and the proposed methods are shown in Figure 15. The red and green lines represent the experimental results of the traditional method and the method proposed in this paper, respectively. It can be seen that the maximum value of the torque jitter is 20.3 Nm at the moment of method switching, and the percentage of torque jitter amplitude is 11.3%. Meanwhile, in the experimental results of the proposed method, the

flag remained constant at 1, and the torque waveform showed minimal jitter, with the maximum jitter being only 4.6 Nm.

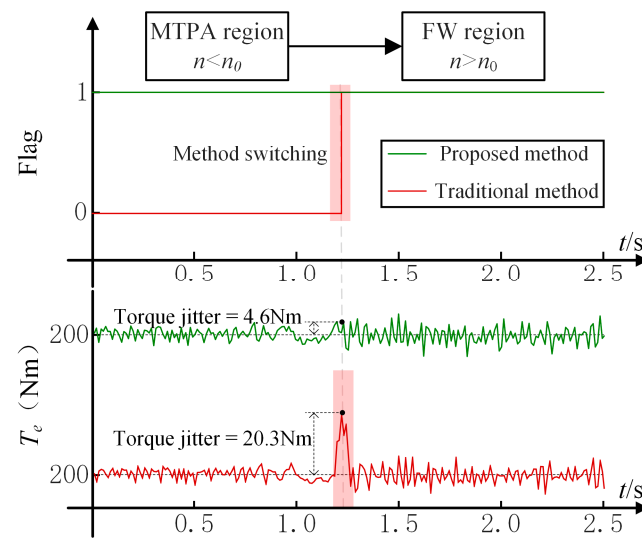


Figure 15. The controllable range of traditional method and the proposed method (200 Nm).

5.2. Experimental Results of Dead-Time Effect Compensation Method

In order to verify the correctness of the proposed method in solving the problem of dead-time effects, a series of comparative experiments was designed. The parameters of the tested motor are shown in Table 3, and the experimental bench is shown in Figure 13. The percentage of dead time to PWM cycle is 3%. The tested motor is set to torque mode, and the target torque changes from 0 Nm to 60 Nm and 100 Nm at $t = 2$ s. The towing motor is set to speed mode, with a motor speed of 5500 rpm.

The comparative experimental results when the target torque is equal to 0.6 times of the rated torque (60 Nm) are shown in Figure 16a. From top to bottom are the waveforms of modulation index, d -axis current, and q -axis current. The black line, red line, and green line represent the reference value, experimental results before, and after compensation. Firstly, focus on the reference value. Within 0~2 s, the torque is equal to 0, so i_{dq} is equal to 0, and the current operating point is located at the coordinate origin. At $t = 2$ s, the torque changes to 60 Nm, the reference modulation index MI^* changes to 0.9, and the optimal current operating point changes to (−50 A, 115 A). Before compensation, the actual modulation index is only 0.84 ($\Delta MI = 0.06$), with a deviation of 7.1% from MI^* . And the actual current operating point is (−81 A, 109 A), with deviations of 62.0% and 5.5% from the optimal point. The proposed compensation method eliminates the influence of the dead-time effect through modulation index compensation. The actual modulation index is 0.896 ($\Delta MI = 0.004$), with a deviation of only 0.4% from MI^* , which is 6.7% lower than that before compensation. And the actual current operating point is (−52 A, 113 A), with deviations of only 4.0% and 1.7% from the optimal point, which is 58% and 3.8% lower than that before compensation.

The comparative experimental results when the target torque is equal to 1 times of the rated torque (100 Nm) are shown in Figure 16b. Note that the comparison results of modulation index are consistent with Figure 16a, proving that the effect of dead time on modulation index is independent of torque. It is consistent with the theoretical derivation mentioned earlier. Before compensation, the actual current operating point is (−106 A, 125 A), with deviations of 32.5% and 10.7% from the optimal point. After compensation, the actual current operating point is (−83 A, 136 A), with deviations of only 3.8% and 2.9% from the optimal point, which is 28.7% and 7.8% lower than that before compensation.

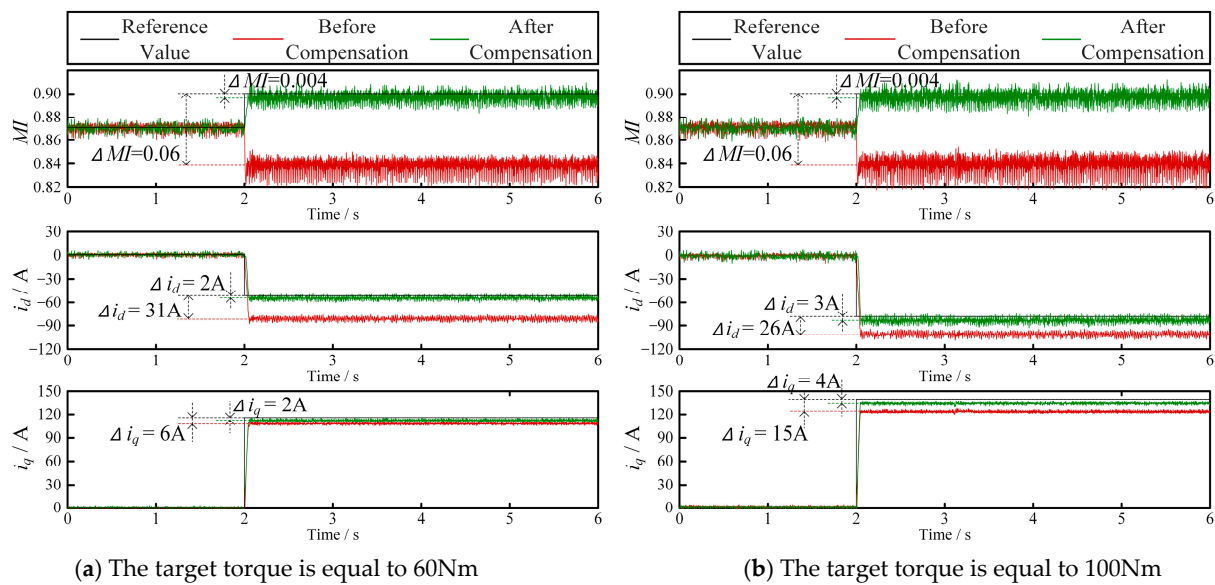


Figure 16. The comparative experimental results before and after compensation.

In order to verify the generalizability and correctness of the proposed compensation method under different voltage modulation indexes, a set of comparative experiments was designed. The reference modulation indexes in the experiment are equal to 0.90, 0.92, 0.94, 0.96, 0.98, and 1, respectively. The quantitative comparison experiment results of the actual modulation index and motor efficiency before and after compensation are shown in Figure 17. The red and green dotted lines represent the actual modulation index before and after compensation, respectively. And the red and green bar charts represent the motor efficiency before and after compensation, respectively. Figure 17a,b show the experimental results with torques equal to 60 Nm and 100 Nm. It can be seen that the actual modulation index after compensation are all greater than those before compensation, with a maximum increase of 6.7%. And the motor efficiency after compensation is higher than that before compensation, with a maximum increase of 2.3%. Therefore, the generalizability and correctness of the proposed compensation method have been demonstrated.

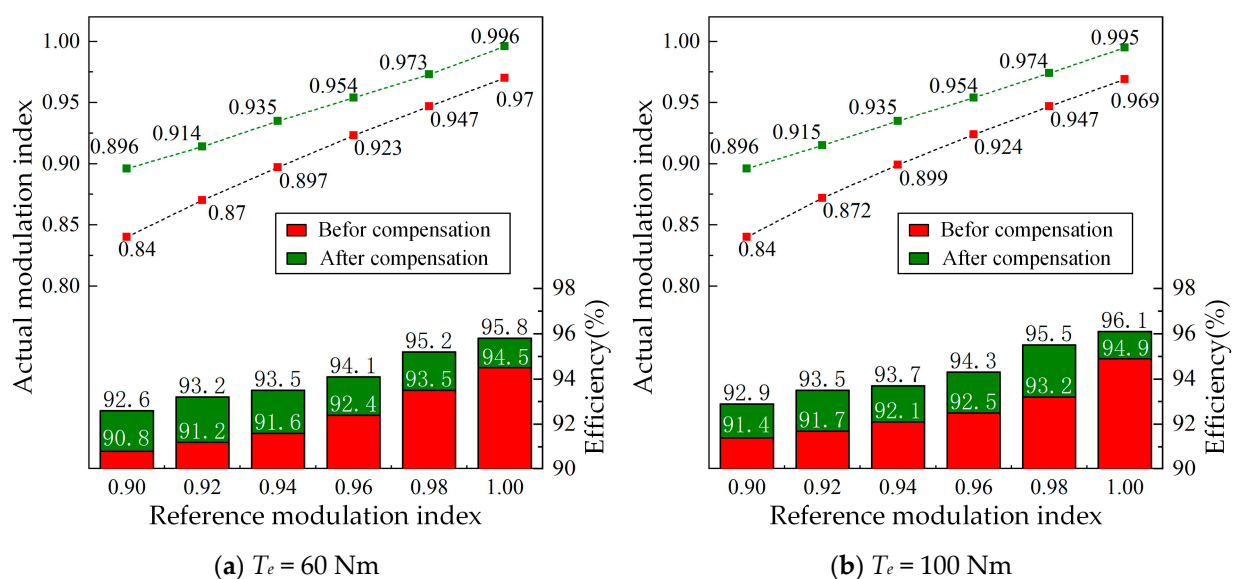


Figure 17. The quantitative comparison experiment results of the actual modulation index and motor efficiency before and after compensation.

6. Conclusions

In this paper, we propose a global-speed-region VAC method considering dead-time effect compensation to address the algorithm switching problem and the dead-time effect problem in the traditional VAC method. The main conclusions are summarized as follows:

- (1) A global-speed-region VAC method. In this article, a global-speed-region method is proposed, which expands the application area of the VAC method to the entire speed region, and solves the problem of torque jitter generated by the traditional VAC method during the transition between low-speed and high-speed ranges;
- (2) A new compensation method for the dead-time effect. In this article, a new dead-time effect compensation method is proposed based on modulation index transformation, in which the current polarity judgment is not required. More importantly, the current operating points are returned to the optimal points, reducing the additional energy loss caused by the dead-time effect.

The proposed method is mainly applicable to motors with wide operating conditions and high output power, such as the drive motors in electric vehicles. The traditional DCR method can be replaced by the global-speed-region VAC method proposed in this article. The target torque analyzed by the accelerator pedal signal can be directly converted into the target voltage through this method, and a stable modulation index can reduce the output torque jitter of the electric power system. Additionally, the proposed method can be applied to both low-speed and high-speed regions, eliminating the torque jitter caused by traditional algorithm switching. Meanwhile, the traditional dead-time effect compensation method can be replaced by the modulation-index-transformation-based dead-time effect compensation method proposed in this paper, which optimizes the compensation effect, improves the control efficiency, and thereby increases the driving range of electric vehicles. Meanwhile, some motor parameters were used in the calculation of the proposed method, so the robustness is somewhat poor, which is something we need to continue researching and improving in the future.

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Conflicts of Interest: The authors declare no conflicts of interest.

Appendix A

The stator voltage equations of PMSM are given as follows:

$$u_d = R_s i_d + L_d \frac{di_d}{dt} - \omega_e L_q i_q \quad (A1)$$

$$u_q = R_s i_q + L_q \frac{di_q}{dt} + \omega_e L_d i_d + \omega_e \psi_f \quad (A2)$$

where u_d and u_q are the dq -axes voltage; i_d and i_q are the dq -axes current; L_d and L_q are the dq -axes inductance; R_s is the stator resistance; ψ_f is the flux linkage; and ω_e is the electrical rotation speed.

And the torque equation of PMSM is as follows:

$$T_e = \frac{3}{2} N_p (\psi_f + (L_d - L_q) i_d) i_q \quad (A3)$$

where T_e is the torque and N_p is the number of poles.

Ignoring the voltage occupied by the stator resistance ($R_s i_d$ and $R_s i_q$) and the current differential terms ($L_d \frac{di_d}{dt}$ and $L_q \frac{di_q}{dt}$), the steady-state voltage equation of PMSM is as follows:

$$u_d = -\omega_e L_q i_q \quad (A4)$$

$$u_q = \omega_e L_d i_d + \omega_e \psi_f \quad (A5)$$

In order to obtain the quantitative relationship between the voltage angle and the output torque, the dq -axes voltages need to be written in polar coordinates.

$$u_d = -u_{max} \sin \theta \quad (A6)$$

$$u_q = u_{max} \cos \theta \quad (A7)$$

where u_{max} is the maximal voltage amplitude ($u_{max} = 2MIu_{dc}/\pi$); u_{dc} is the DC bus voltage; and θ is the voltage angle between the voltage vector and the q -axis.

Appendix B

The equation of the MTPA curve is as follows:

$$\left(i_d - \frac{i_b}{2}\right)^2 - (i_q)^2 = \left(\frac{i_b}{2}\right)^2 \quad (A8)$$

where $i_b = \frac{\psi_f}{(\rho-1)L_d}$.

By combining the torque Equation (A3) and the MTPA curve Equation (A8), we obtain the following:

$$\left(i_d - \frac{N}{2}\right)^2 - \left(\frac{2T_e}{3n_p[\psi_f - L_{err}i_d]}\right)^2 = N^2 \quad (A9)$$

where $N = \frac{\psi_f}{(\rho-1)L_d}$, $L_{err} = L_q - L_d$.

By opening the square term in Equation (A9), we can obtain a univariate quartic equation about i_d as follows:

$$4KH i_d^4 - (8HM + 4HKN) i_d^3 + (4HL + 8HMN + HK - 4GHK) i_d^2 - (4NL + 2MN^2 - 8GHM) i_d + (LN^2 - 4GHL - 16T_e^2) = 0 \quad (A10)$$

where $G = \left(\frac{N}{2}\right)^2 = \left(\frac{\psi_f}{2(\rho-1)L_d}\right)^2$, $H = 9n_p^2$, $K = L_{err}^2 = (L_q - L_d)^2$, $L = \psi_f^2$, $M = \psi_f L_{err} = \psi_f (L_q - L_d)$.

Equation (A10) only contains variables i_d and T_e , so i_d can be written as a function of T_e , i.e., $i_d = f(T_e)$. i_d can be obtained using the standard quadratic root formula.

$$i_d = f(T_e) = i_{d3} = -\frac{b}{4a} - \frac{1}{2}\sqrt{V + \frac{1}{3a}Y} + \frac{1}{2}\sqrt{-4\left(\frac{1}{2}\sqrt{V + \frac{1}{3a}Y}\right)^2 - W + \frac{X}{\frac{1}{2}\sqrt{V + \frac{1}{3a}Y}}} \quad (A11)$$

where $P = 2c^3 - 9bcd + 27b^2e + 27ad^2 - 72ace$, $S = c^2 - 3bd + 12ae$, $V = \frac{3b^2 - 8ac}{12a^2}$, $W = \frac{8ac - 3b^2}{4a^2}$, $X = \frac{b^3 - 4abc + 8a^2d}{8a^3}$, $Y = \sqrt[3]{\frac{P + \sqrt{P^2 - 4S^3}}{2}} + \frac{S}{\sqrt[3]{\frac{P + \sqrt{P^2 - 4S^3}}{2}}}$.

Substituting Equation (A11) into Equation (A3), i_q can be obtained.

$$i_q = g(T_e) = \frac{2T_e}{3n_p[\psi_f + (L_d - L_q)i_{d3}]} \quad (A12)$$

Substituting i_{dq} into the (A4) and (A5), the u_{dq} can be obtained.

$$\begin{cases} u_d = -\omega_e L_q g(T_e) \\ u_q = \omega_e L_d f(T_e) + \omega_e \psi_f \end{cases} \quad (\text{A13})$$

The amplitude of the voltage vector corresponding to the intersection of the ETC curve and the MTPA curve is as follows:

$$|\mathbf{u}_{mtpa}| = \sqrt{(\omega_e L_q g(T_e))^2 + (\omega_e L_d f(T_e) + \omega_e \psi_f)^2} \quad (\text{A14})$$

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